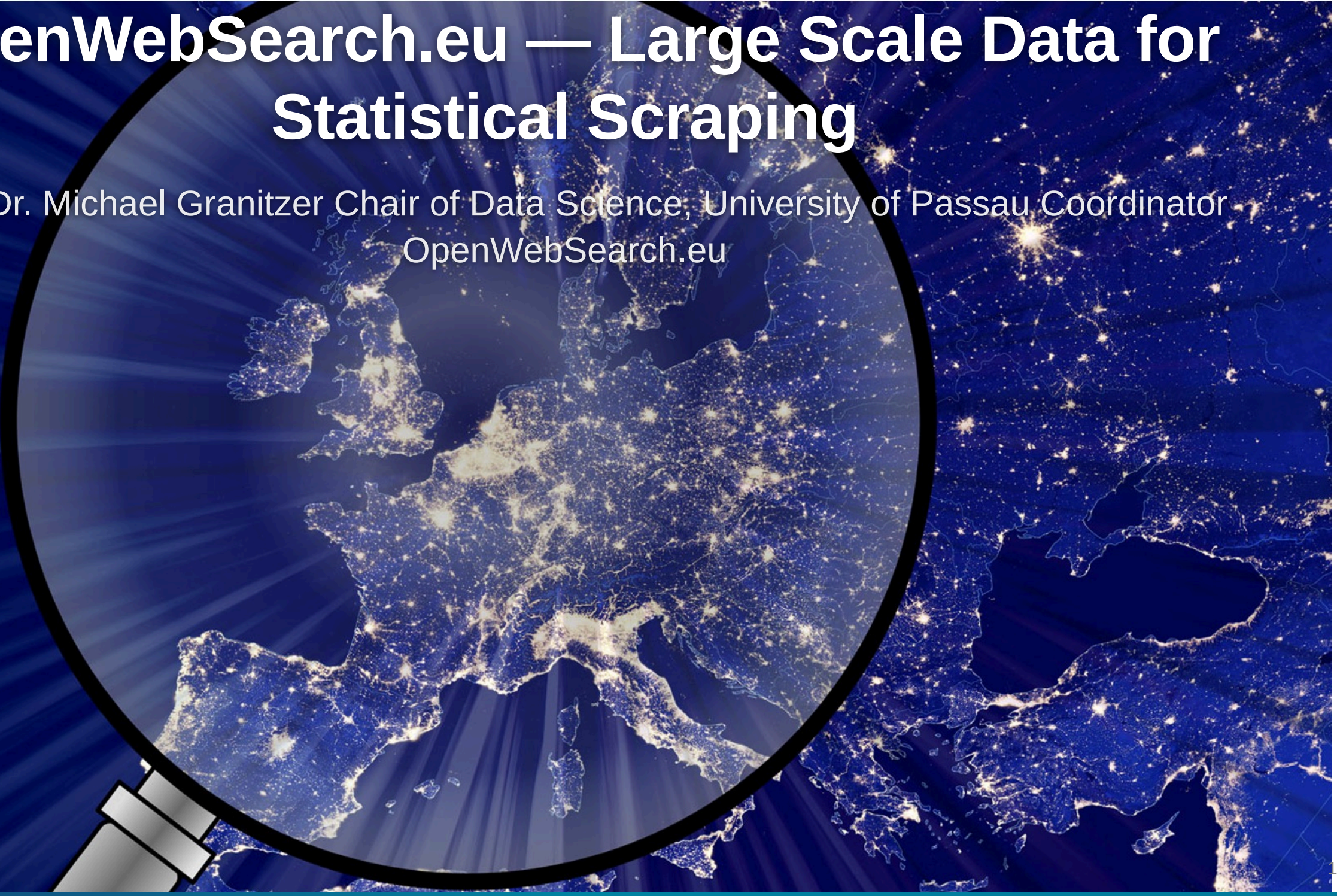


# OpenWebSearch.eu — Large Scale Data for Statistical Scrapping

Prof. Dr. Michael Granitzer Chair of Data Science, University of Passau Coordinator  
OpenWebSearch.eu



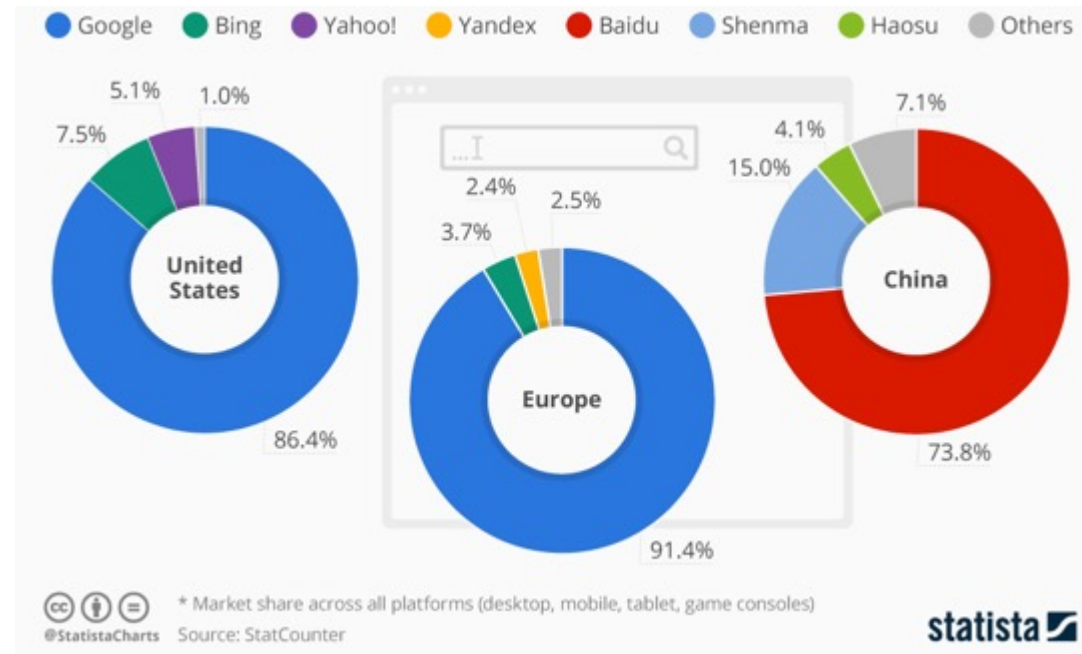


# Web Search Today

A critical resource managed as an oligopoly

## Two properties that don't fit

- A **critical infrastructure** for society, comparable to satellite navigation
- A **market oligopoly**: dominated by a small number of gatekeepers (Google, Microsoft, Baidu, ...)



# But does it matter? Yes — on four dimensions

## User Choice & Democracy

- Single gatekeeper for society's information
- Search rankings can shift voting preferences by **20–72%** (SEME)
- Monopolistic ranking = single point of failure for information quality

## Untapped Richest Information Source

- 60 % of web pages carry structured data
- No open access to web-scale document collections
- Science, research and technology heavily depend on the Web as platform

## Societal & Security Risks

- European OSINT depends on US-controlled infrastructure
- Filter bubbles, algorithmic bias
- Misinformation & Disinformation

## AI & Innovation

- Web data is the fuel for LLMs, RAG, and agentic AI
- Closed indices block reproducible research & open innovation

Being able to navigate the Web is as fundamental as GPS or the power grid

# Challenges

## THE WEB: VOLUME, VELOCITY, VARIETY

Careful: domains, sites, pages, and indexed documents are different units

### VOLUME

**1.15 Billion**

Netcraft site responses

272.6 M domains · Dec 2024 Netcraft survey

**Top 0.1%–1% of sites**

may account for roughly half of all pages  
heavy-tail intuition from large web graph studies

**~ 400 Billion**

documents in Google's search index  
2020 figure cited in US v. Google testimony

*Sites ≠ domains ≠ pages; use exact counts cautiously*

### VELOCITY

**+8.6 M / month**

monthly net change in Netcraft site responses

Dec 2024 versus Nov 2024 Netcraft survey

**~ 3.2 / second**

same monthly net change, averaged across December

**149 ZB (global)**

all digital data created / copied worldwide in 2024  
IDC / Statista global datasphere estimate

*The web changes constantly; crawling is continuous, not one-off*

### VARIETY

**150+ languages**

observed across websites

W3Techs usage survey

**249 CMS Platforms**

CMS platforms identified in the HTTP Archive dataset  
HTTP Archive Web Almanac 2024 (Wappalyzer-based)

**HTML · PDF · JSON-LD · media · APIs**

content types to parse and process

*HTML · PDFs · JS apps · schemas · media · APIs*

Sources: Netcraft Web Server Survey Dec 2024 · large web graph studies (heavy-tail intuition) ·  
US v. Google testimony on ~400B documents (2020 figure) · HTTP Archive Web Almanac 2024 · W3Techs · IDC/Statista



- High infrastructure costs
- Broad range of technical skills required
  - Big Data Infrastructure
  - Natural Language Processing, Image Analysis
  - Web Technologies
- Growing legal uncertainties

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- High infrastructure costs
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  - Natural Language Processing, Image Analysis
  - Web Technologies
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High upfront costs to tap the Web as resources — especially small innovators and researchers are left behind



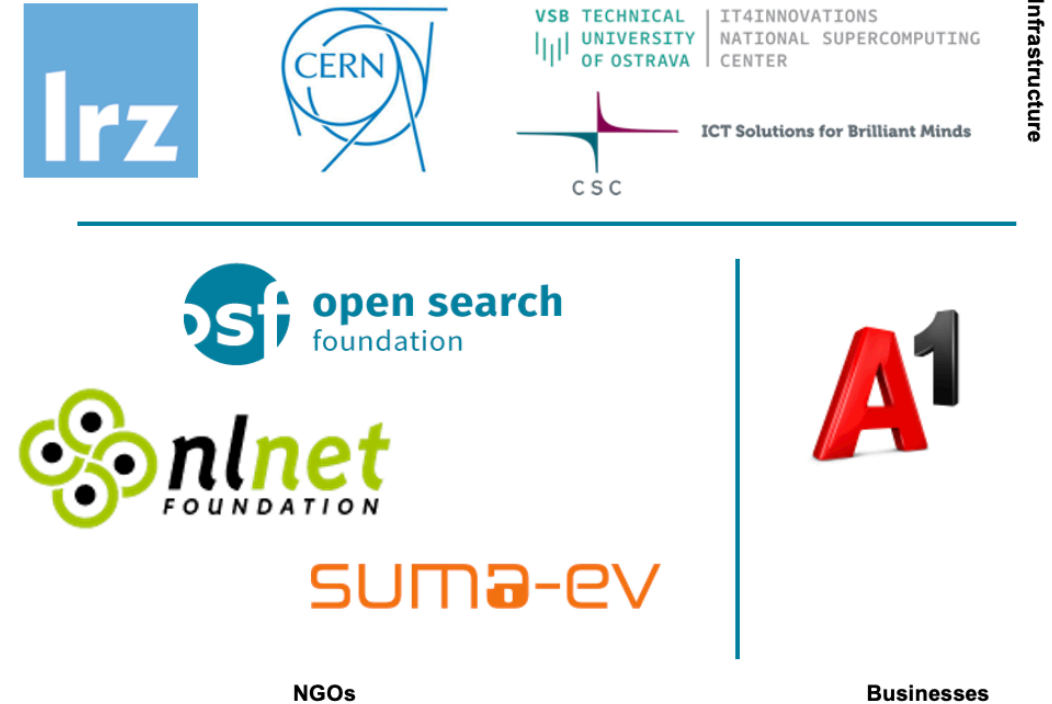
# OpenWebSearch.eu — Our Mission

Building an Open Index of the Web and an open infrastructure to reduce upfront costs for researchers, innovators and companies in tapping the Web as a resource for search, web-analytics and AI.

## Four Objectives

- 1. Open Technology Stack**  
Open-source pipelines for crawling, preprocessing, indexing
- 2. Resource Provision** Building a network of infrastructure providers (EuroHPC / AI Factories)
- 3. Added Value Services**  
Dashboard, tools, data access APIs
- 4. Bootstrapping the Ecosystem** Third-party calls, community building, applications and use-cases

## 14 Core Partners

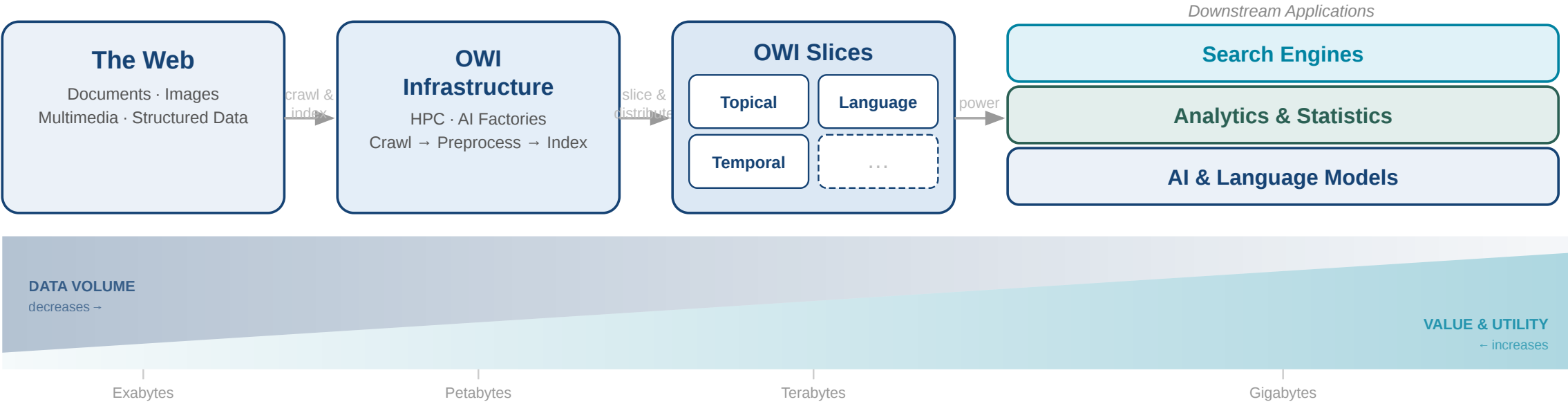


Plus 16 financially supported third-party projects (FSTPss)

# The Open Web Index (OWI)

A **Web Index** = a data structure for fast query-based access and ranking of web documents — the core of every web search engine

**Our Proposition:** A collaboratively created, open and transparent Web Index running on existing HPC computing centres / AI Factories in Europe



Granitzer, Michael, et al. "Impact and development of an Open Web Index for open web search." *Journal of the Association for Information Science and Technology* (2023).

# What is the Open Web Index?

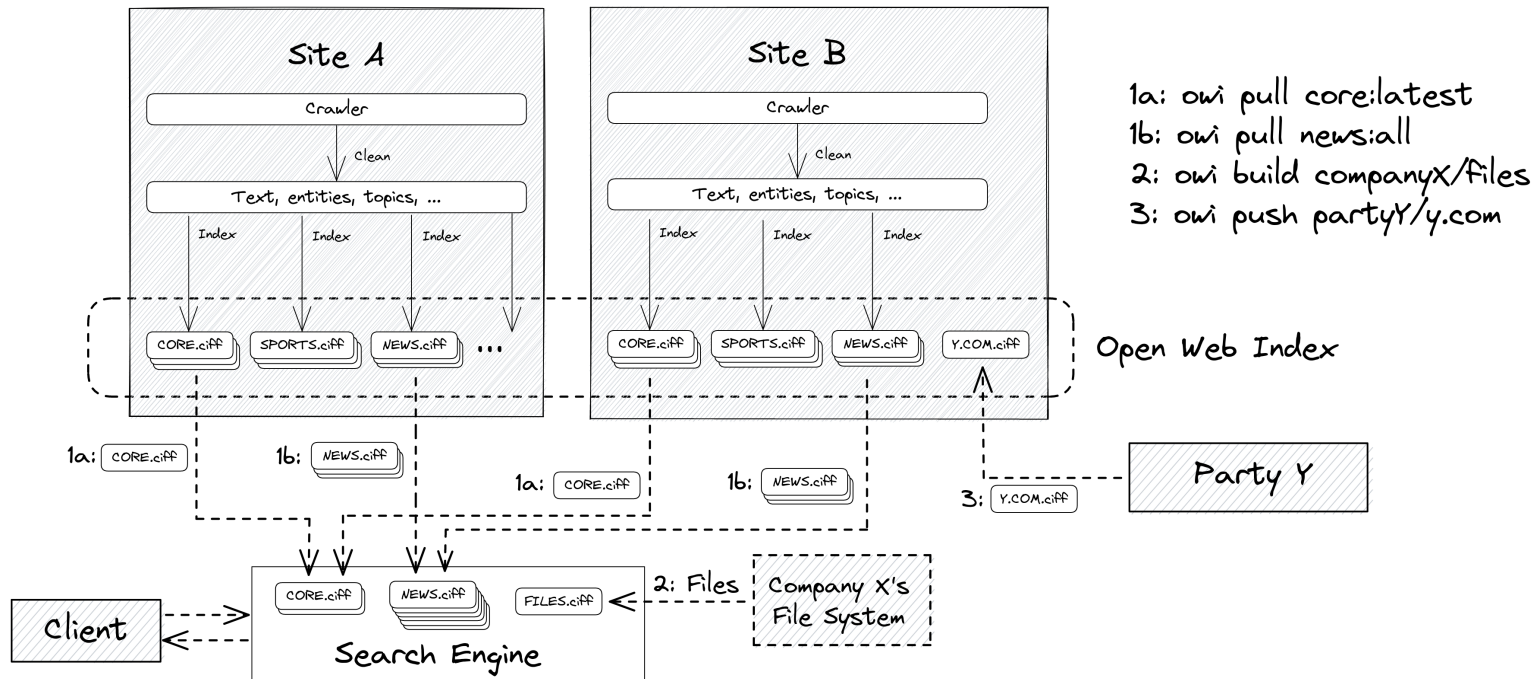
# The Open Web Index

**The Open Web Index** — a federated, daily-published web index built across European HPC sites and consumed as composable shards.



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## Federated production

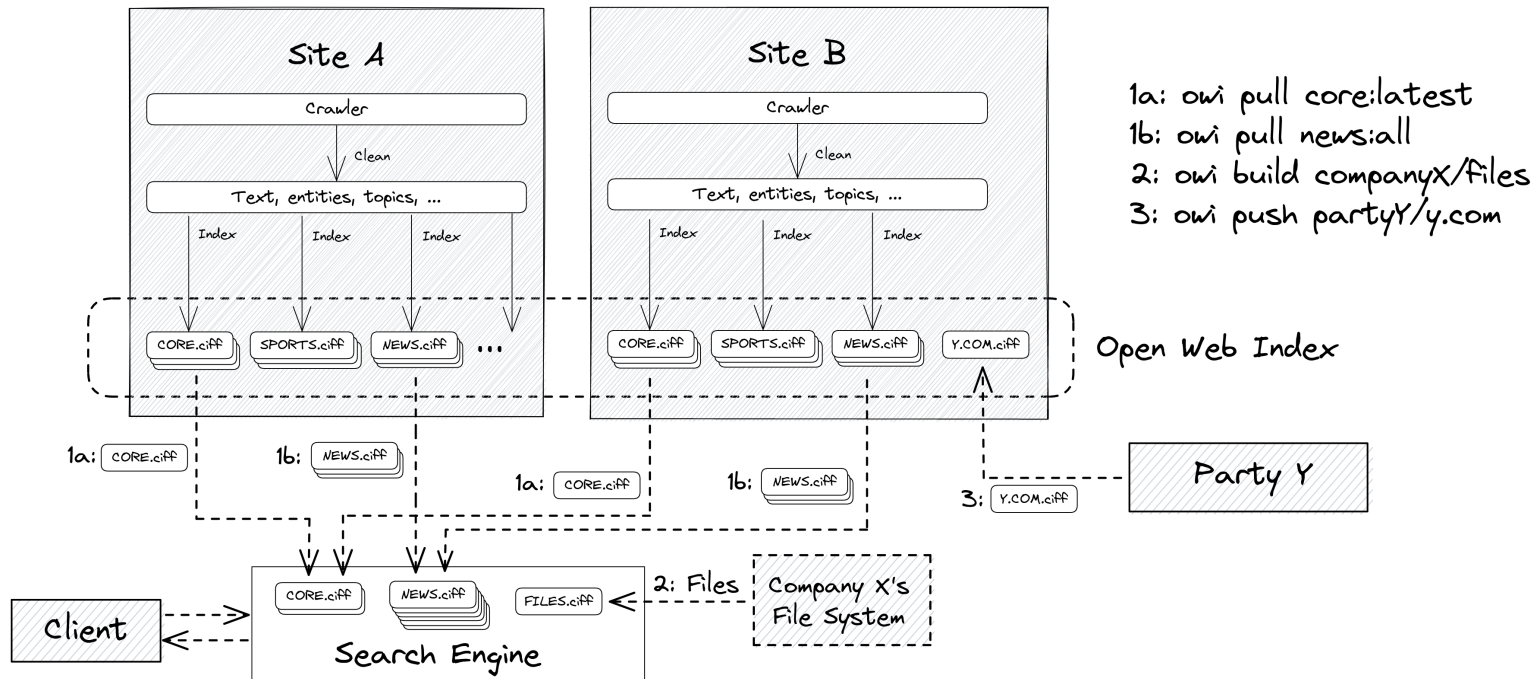
- each HPC site runs the same pipeline independently
- daily shards by date, language, topic

## Three access verbs

- owi pull — download pre-built shards
- owi build — index your own data
- owi push — contribute shards back

# The Open Web Index

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## Federated production

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## Three access verbs

- **owi pull** — download pre-built shards
- **owi build** — index your own data
- **owi push** — contribute shards back

Three output types per daily run — the building blocks of a shard:

# OWI at a Glance — Statistics (Feb 2026)

## Crawling

|                 |                           |
|-----------------|---------------------------|
|                 |                           |
| Pages crawled   | ~39 B total, ~12 B unique |
| Unique hosts    | ~500 M (extrapolated)     |
| Crawling effort | 1,927 machine-days        |
| Avg. throughput | ~20 M URLs / day          |

## Index

|                   |                            |
|-------------------|----------------------------|
|                   |                            |
| Documents indexed | ~10.7 B                    |
| Unique domains    | ~75 M                      |
| Public datasets   | 511 main / 1,511 total     |
| Index size        | 45 TB (main) / 50 TB total |
| Embeddings        | 119 M docs · 884 GB        |

- Top languages:** English (38.9%), German (7.2%), Spanish (6.4%), French (5.7%), Russian (5.2%)
- Multimedia Objects:** Roughly 2-3 quality images and 0.5 Videos per Web-Document
- Comparison:** To meet commercial index size, OWI would need to scale by a factor of 20× — factor 10–50× for multimedia

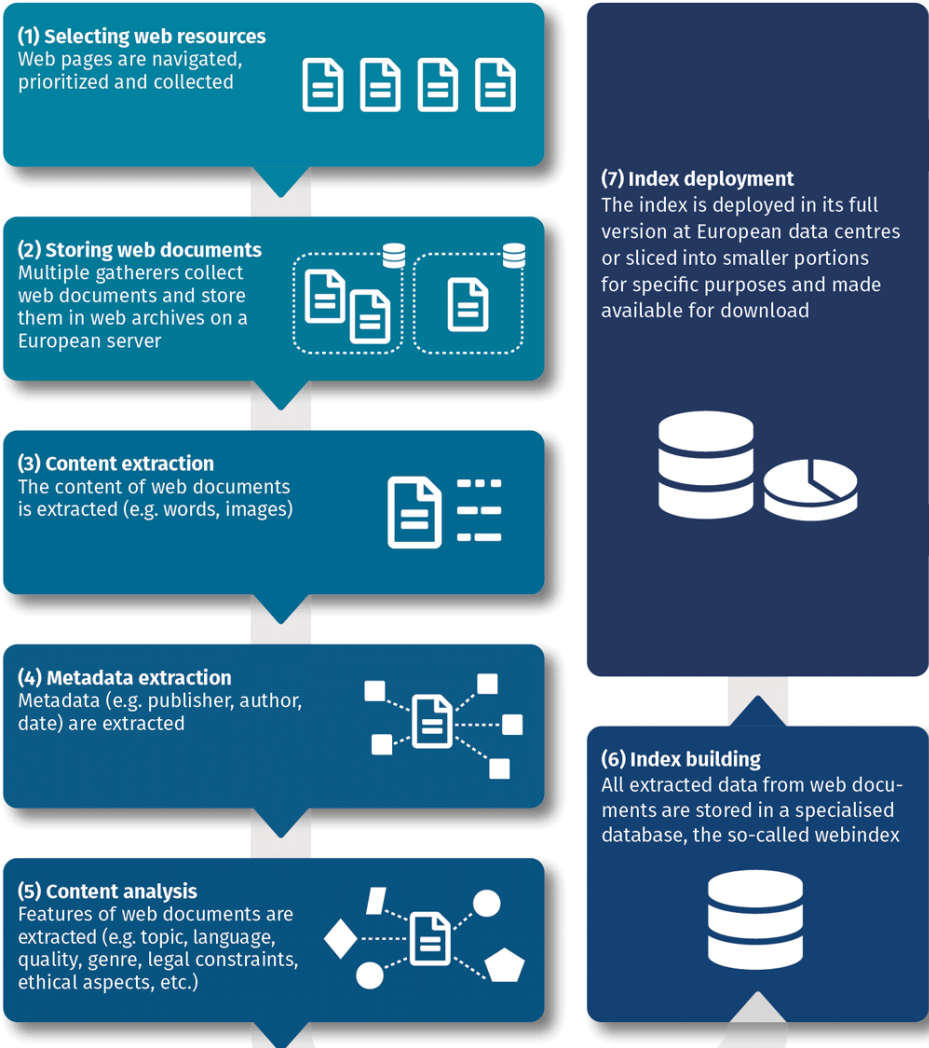


# Building the Open Web Index

# Conceptual Workflow

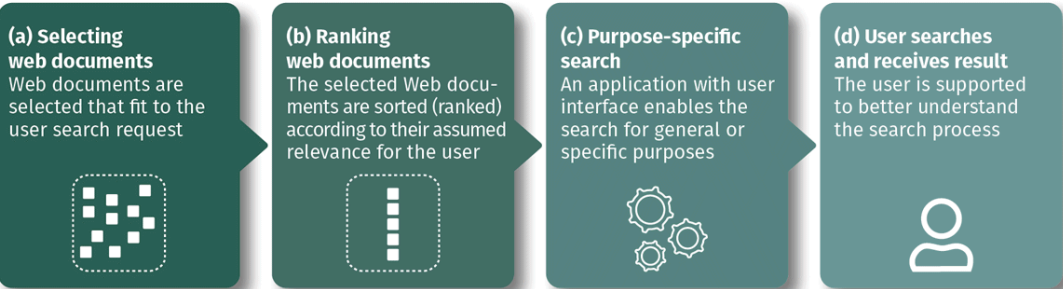
## Index Generation

Web resources are selected and retrieved, their content and metadata are analysed, and all data stored in the index database.



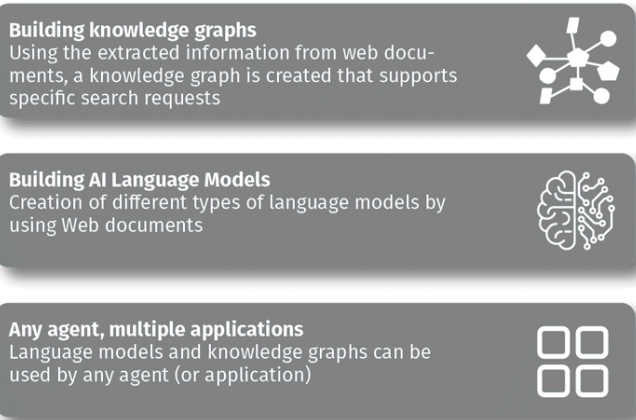
## Search Applications

A user search request will be answered by a search application that makes use of the open web index.



## Data Products

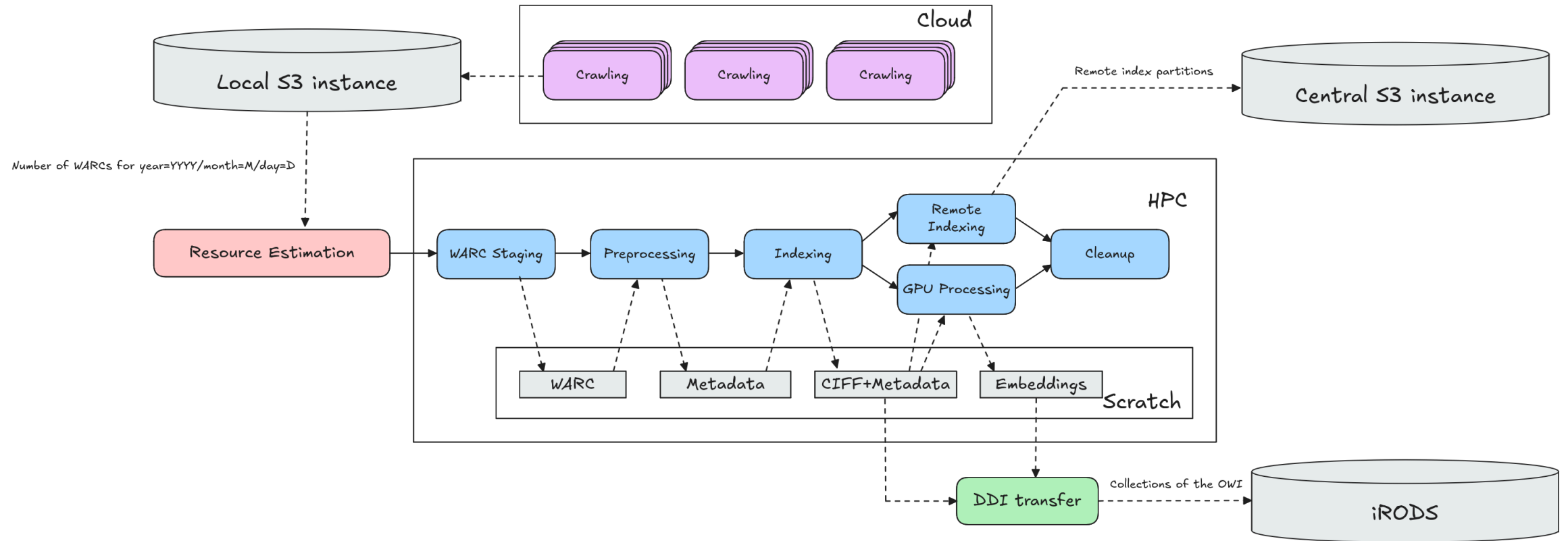
Knowledge representation models will be created using the open web index, in order to be used by any agent and for many applications



...



# Daily Workflow Pipeline



## Crawlig Infrastructure (2026)

- **25 VMs** across IT4I, LRZ, CSC, CERN
- **100 M URLs / day** crawling capacity
- **~1.1 PB** raw data · **~50 TB** index · **810+ public datasets**
- Federated storage: iRODS + S3, orchestrated via LEXIS

## Limitations as of 2026

- TLR 5/6: it is not a product (yet)
- No 24/7 Operations
- Limited funding - minimum engineering / improvement





# Accessing the Open Web Index

# The Dashboard — openwebindex.eu

OWI

Dashboard

SERVICES

Websites

Search

SerCI Search EXTERNAL

Crawling

EXPERIMENTAL

Deep Research

UTILITIES

Parse Preview

Ethics Readiness Check

Impressum

Privacy Statement

Terms of Use

Open Web Search Book

Open WebSearch

Open Web Index

Overview OWI Statistics OWI Datasets OWI Corpora OWI Models

## The Open Web Index

Your daily dose of web data!

Welcome to the Open Web Index (OWI)- your way to slice and dice the Petabytes in the Web to your needs. Our mission is to contribute to a fair, open, diverse and free Web by providing an open Index of the Web for you to use. This dashboard gives an overview over this [Open Web Index](#) and provides some prototype services on top of the OWI.



### What can you find here?

Discover our comprehensive suite of tools and services designed to help you explore and analyze web data.

**OWI Statistics**  
Get an overview of the data we offer

**OWI Datasets**  
Browse through available datasets

**OUR Search**  
Search through URLs and Titles

**Documentation**  
Access tutorials and detailed documentation

**Expect Bugs:** While we try to provide as many valuable datasets and services as possible, please always remember that this is still a research project. So we cannot guarantee that everything works perfectly fine and we cannot take any responsibility if something is not working as intended. Bugs are to be expected, but if you encounter some, please get in touch with us so that we can improve.

## Central entry point for the Open Web Index [openwebindex.eu](https://openwebindex.eu)

- **OWI Statistics** — crawl volume, index size, language distribution, dataset timeline
- **OWI Datasets** — browse, filter and download index shards (by DC, date, format, language)
- **Search** — demo content search + SERCI integration

424 registered users · 17,000+ unique visitors · 70,000+ page views

# Dashboard — Statistics & Datasets



- ### OWI Statistics (04/26)
- **1,389 TB** crawled · **281** WARC datasets · **1,696** public datasets · **35.09 TB** index
  - Language distribution chart (English 38%, German 8%, French 6%, ...)
  - Crawl volume over time per data center + dataset collection distribution by type
  - Datasets over time

# Dashboard — Statistics & Datasets

### Available Datasets

The data shows the datasets available for download (i.e. public owi datasets) or upon request (mostly project private warc datasets) by using our [LEXIS Platform](#) or our [OWI Command Line Tool](#). A dataset is a temporal slice, most often a single day, of crawled (warc), preprocessed and indexed data (owi).

All systems are online

All components are functioning properly. Data repositories and services are accessible.

Last Updated: 12/04/26, 15:36

| SYSTEMS | DESCRIPTION | COMPONENTS |            |         | STATUS |        |
|---------|-------------|------------|------------|---------|--------|--------|
| IT4I    | IT4I        | connected  | Repository | Project | Public | ONLINE |
| LRZ     | LRZ         | connected  | Repository | Project | Public | ONLINE |

Filter datasets...

All Collections

All Datacenters

All Resources

Public

Sort by: Collection Date

OWI - Open Web Index

05/04/2026 → 05/04/2026

LEGAL IT4I PUBLIC

204.5 MB

253 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fcb68e42-3191-11f1-8cbe-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

MAIN IT4I PUBLIC

24.3 GB

1,065 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faba586e-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

CURLIE\_FULL IT4I PUBLIC

2.1 GB

365 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fabf3050-3191-11f1-af7e-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

LICENSES IT4I PUBLIC

63.6 MB

163 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

05/04/2026 → 05/04/2026

GPU IT4I PUBLIC

3.6 GB

374 files

All OWI data crawled between 2026-04-05 and 2026-04-05 at CSC2

owilix remote pull all/internalID=faf9d060-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

GPU IT4I PUBLIC

40.8 MB

6 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

LICENSES IT4I PUBLIC

275.8 KB

2 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=fac99de2-3191-11f1-b11f-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

CURLIE\_FULL IT4I PUBLIC

5.3 MB

8 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=faf9d060-3191-11f1-9f25-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

OWI - Open Web Index

03/04/2026 → 03/04/2026

MAIN IT4I PUBLIC

108.4 MB

73 files

All OWI data crawled between 2026-04-03 and 2026-04-03 at CSC2

owilix remote pull all/internalID=2f8f678c-2fd5-11f1-8ad8-3e9ca2b0f439

VIEW STATISTICS

View Dataset License

Open in LEXIS Portal

## OWI Datasets Tab

- Status of repositories
- Filter by datacenter, format (WARC / Parquet / CIFF / Embeddings), language, date
- Dataset cards with download links — LEXIS Portal or owilix CLI
- Parquet file preview directly in the browser

# Download Tool: owilix

## owilix — The OWI Command-Line Tool

Git-like versioning for index datasets — pull, build, push:

```
owilix remote pull all:latest/collectionName=main
owilix local ls all:latest
```

Slice by: **domain**, **language**, **topic** (Curly), **sitelist**,  
**structured data**, **outlinks**

Authentication via B2ACCESS / Keycloak (LEXIS SSO)

## Remote Index — Query first, then Download

- Parquet partitions directly queryable via **DuckDB** or **Apache Spark**
- Full-text search
- SQL access: filter by language, domain, date — no full-shard download needed
- Enables search + filtering + download (but still in alpha phase)

## OWILIX - Open Web Index CLI

version 5.3.1 python 3.11+ license Apache 2.0

A command-line tool for managing [OpenWebIndex](#) datasets across local and remote repositories.

### Quick Start

#### One-Line Install (Recommended)

```
curl -fsSL https://openwebindex.net/owilix/install.sh | sh
```

<https://opencode.it4i.eu/openwebsearcheu-public/owi-cli/>

```
➔ ~ owilix remote search "michael granitzer" --limit 10
Searching remote index: terms=['michael', 'granitzer'] language=eng representation=main_content mode=OR limit=10
Found 10 results in 113.4s
Search results (10 results in 113.4s)
```

| score | date       | url                                                          |
|-------|------------|--------------------------------------------------------------|
| 3.088 | 2025-12-30 | https://easychair.org/smart-program/ICADL2024/person16.html  |
| 3.010 | 2026-02-19 | https://www.wolframalpha.com/input/?i=michael                |
| 3.010 | 2026-01-19 | https://www.michaelmoennich.com/                             |
| 3.010 | 2025-12-08 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-01-05 | http://www.michaelmoerk.dk/                                  |
| 3.010 | 2026-01-22 | https://michaelkoenig.de/                                    |
| 3.010 | 2026-02-18 | https://www.sktthemes.org/forums/users/dedicationpt/replies/ |
| 3.010 | 2026-02-26 | https://www.abeldiaz3.com/tag/michael/                       |
| 3.010 | 2026-02-20 | https://discourse.psychopy.org/u/michael                     |
| 3.010 | 2026-02-27 | https://thedigitalgobag.com/author/michael/                  |

# The OpenWebSearch Book

Comprehensive Documentation at [openwebindex.eu/book](https://openwebindex.eu/book)

- Covers the full OWI ecosystem:
- Getting started with owilix
  - Building vertical search engines with MOSAIC
  - Data formats: CIFF, Parquet, embeddings
  - Pipeline architecture and HPC deployment
  - Legal & ethical guidelines for index users
- Published as an open, continuously updated online resource — aligned with deliverables and community feedback.

# Access Modes at a Glance

| Tool                      | Mode           | Best for               |
|---------------------------|----------------|------------------------|
| <a href="#">Dashboard</a> | Browser        | Explore, monitor       |
| <a href="#">owilix</a>    | CLI            | Download, slice, build |
| Remote index              | DuckDB/SQL     | Query + download       |
| <a href="#">MosaicRAG</a> | Conversational | RAG & chat over OWI    |

All software, data formats and access tools are **open source**





# Using the Open Web Index — Applications

# MosaicRAG — Conversational Search on the OWI

**Modular RAG framework** extending MOSAIC with LLM-based capabilities and dense retrieval (WEBIS / TU Graz)

- **Two retrieval paths:** inverted index (CIFF) + dense retrieval
- **4 reranking strategies:** TF-IDF, Embedding cross-encoder, Tournament-style (LLM pairwise), Group-style (LLM window)
- **LLM backends:** self-hosted on-premise models for summarization
- **Conversational search:** follow-up questions about retrieval results; dual-window UI (search results + chat)
- **Workflow:** define retrieval pipeline → issue query → converse about results

## Key Features

- Relevance marking — LLM highlights relevant passages within documents
- Summarization — user-provided instructions applied per document
- Domain-specific indices: science, health, arts (8M+ docs)
- Fully open-source modular pipeline

## Data Pipeline

OWI daily datasets → CIFF (inverted index) + Parquet (metadata) + dense embeddings (since Dec 2025) → MosaicRAG



Search result list

Conversational search

Knowledge graph

Logs

Retrieval pipeline

Reset chat

Language model  
gemma2

Text column  
full-text

Add step

Reset everything

Get link

list universities in Austria

Here are some universities in Austria, categorized for clarity:

Public Universities:

- **University of Vienna (Universität Wien):** The largest and oldest university in Austria, renowned for its humanities, social sciences, and natural sciences programs.
- **Technical University of Vienna (Technische Universität Wien):** Focuses on engineering, technology, natural sciences, architecture, and mathematics.
- **University of Graz (Karl-Franzens-Universität Graz):** Known for its strong humanities, social sciences, law, medicine, and technical science faculties.
- **University of Innsbruck (Leopold-Franzens-Universität Innsbruck):** Offers programs in a wide range of disciplines, with strengths in the natural sciences, economics, and law.
- **Johannes Kepler University Linz (JKU Linz):** Specializes in engineering, natural sciences, social sciences, economics, and law.

MosaicDataSource

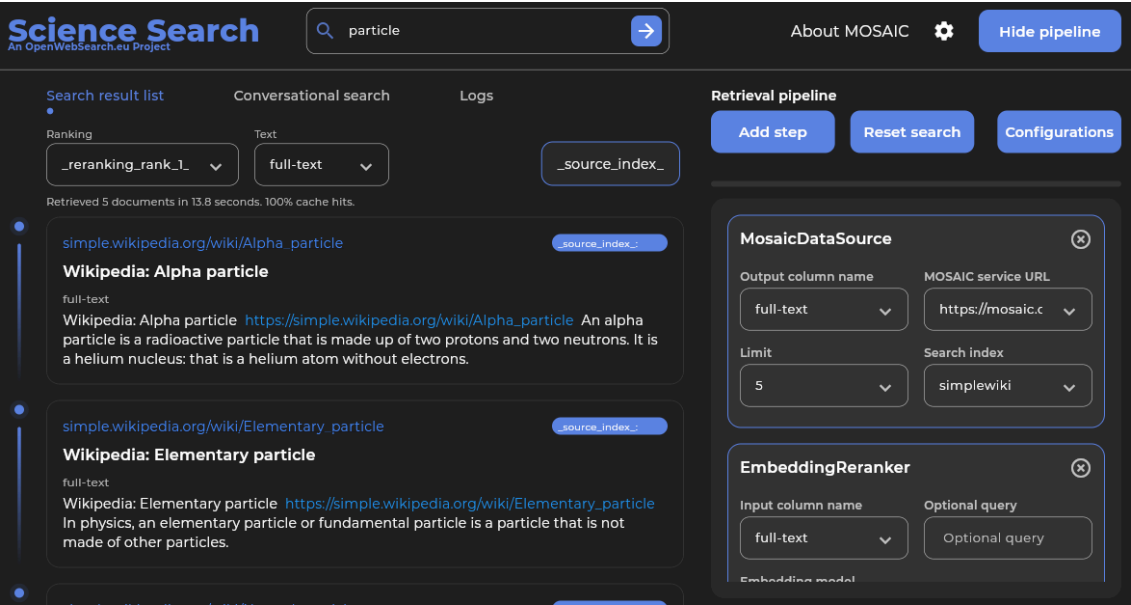
|                    |                    |
|--------------------|--------------------|
| Output column name | MOSAIC service URL |
| full-text          | https://mosaic.fe  |
| Limit              | Search index       |
| 20                 | unis-austria       |

Demo: [mosaicrag.ows.eu](https://mosaicrag.ows.eu) · [mosaic.ows.eu](https://mosaic.ows.eu)

# Mosaic Search Applications · Earth Observation Search

## Mosaic Vertical Search (TU Graz)

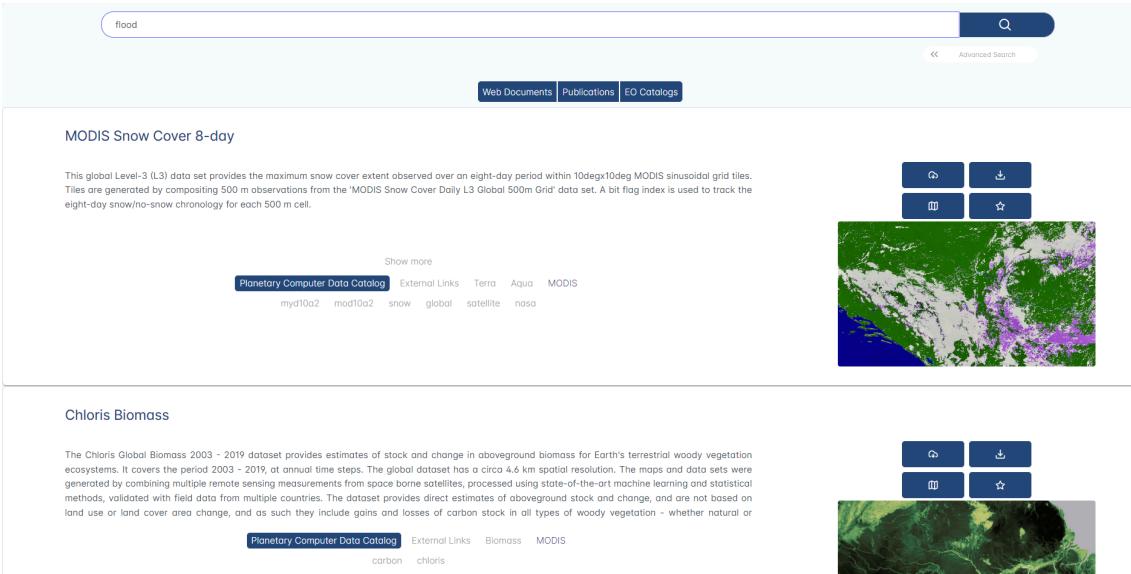
Modular framework for building vertical search engines on OWI slices — domain-specific indices for science, health, arts (8M+ docs)



Demo: [mosaic.ows.eu](https://mosaic.ows.eu) · [mosaic.ows.eu/demos](https://mosaic.ows.eu/demos)

## Earth Observation Search (DLR / Radboud)

RAG-based QA over scientific papers, datasets & web content; geo-contextualized multi-genre search on interactive maps; KG with 2M+ nodes (OpenAlex, PANGAEA, DLR Geoservice)



Demo: [eo-rag-demo.2.rahtiapp.fi](https://eo-rag-demo.2.rahtiapp.fi) · [ows-demo.2.rahtiapp.fi](https://ows-demo.2.rahtiapp.fi)

# Multilingual Scientific Index · Restaurant Recommender

## Multilingual Scientific Index (CSC)

Enriches Finland's Research.fi portal with OWI web data — Finnish & English science content classified by type (publication, blog, event); embedded directly in national research portal

### Energy-centric optimization of large-scale additive manufacturing deployment for sustainability through electrification

|                                     |  |                                                                                                                                                                                                                                                                                              |                                                         |           |
|-------------------------------------|--|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------|-----------|
| Rahoitetun hankkeen kuvaus          |  | The employment of additive manufacturing (AM), commonly called 3D printing, in the industrial sector has been growing for many years. As the technology matures, it is expected that its use will become widespread and substitute other established forms of production. This project is... |                                                         |           |
| 1                                   |  | Näytä enemmän                                                                                                                                                                                                                                                                                |                                                         |           |
| Aloitusvuosi                        |  | 2026                                                                                                                                                                                                                                                                                         |                                                         |           |
| Päätymisvuosi                       |  | 2029                                                                                                                                                                                                                                                                                         |                                                         |           |
| Myönnetty rahoitus                  |  | Humberto Almeida Junior                                                                                                                                                                                                                                                                      | Lappeenrannan–Lahden teknillinen yliopisto LUT          | 305 350 € |
| Rooli Suomen Akatemian konsortiossa |  | Partneri                                                                                                                                                                                                                                                                                     |                                                         |           |
| Muut osapuolet                      |  | Partneri                                                                                                                                                                                                                                                                                     | Aalto-yliopisto (372725)                                | 298 731 € |
|                                     |  | Johtaja                                                                                                                                                                                                                                                                                      | Lappeenrannan–Lahden teknillinen yliopisto LUT (372723) | 395 919 € |
| Rahoittaja                          |  | Suomen Akatemia                                                                                                                                                                                                                                                                              |                                                         |           |
| Rahoitusmuoto                       |  | Suunnattu akatemiahanke                                                                                                                                                                                                                                                                      |                                                         |           |
| Haku                                |  | 2025 Tulevaisuuden kestävien energiaratkaisujen tutkimuksen haku 2025                                                                                                                                                                                                                        |                                                         |           |
| Muut tiedot                         |  |                                                                                                                                                                                                                                                                                              |                                                         |           |
| Rahoituspäätöksen numero            |  | 372724                                                                                                                                                                                                                                                                                       |                                                         |           |
| Tieteenalat                         |  | Kone- ja valmistustekniikka                                                                                                                                                                                                                                                                  |                                                         |           |
| Tutkimusalat                        |  | Kone- ja valmistustekniikka                                                                                                                                                                                                                                                                  |                                                         |           |

### Myöntöön liittyvä verkkosivu

Rahoitusmyöntöön ei ole tarjolla linkkejä tässä portaalissa

### Viittaukset muissa palveluissa

Viittauksia yhteensä: 20



### Hakutulokset 1 - 5

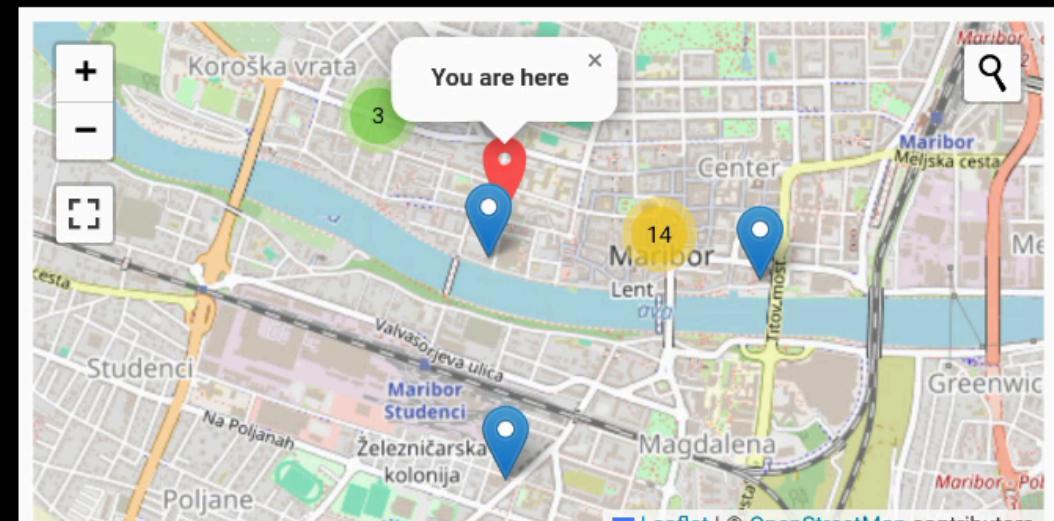
- <https://www.industryweek.com/...>  
3 Ways Additive Is Shaping the Factory of the Future | IndustryWeek
- <https://www.industryweek.com/...>  
These Industries are the Future of Additive Manufacturing | IndustryWeek
- <https://global-sci.com/...>  
Global Science Press: A Review of Process Optimization for Additive Manufacturing Based on Machine Learning
- <https://www.engineering.com/...>  
Customized Production Will Become Scalable with 3D Printing - Engineering.com
- <https://axial.acs.org/...>  
Welcome the Newest Associate Editors of ACS Publications Journals: Q3 2018 | ACS Publications Chemistry Blog

< 1 2 3 4 >

## Restaurant Recommender (A1 Slovenia)

Privacy-preserving mobile app; on-device personalization via query imputation; LLM-extracted restaurant features from OWI; 5-month pilot with 214 users — overall satisfaction 4.2 / 5

# Restaurant Recommender



Or search for restaurants in 1 km radius

Select City ▼

Bascarsija grill food



## Baščaršija

[Visit Restaurant Page](#)

Poshtna ulica 8, Maribor 2000 Slovenia 📞 +386 2 250 63 59

Cuisines: Barbecue; European; Grill; Slovenian; Eastern European Meals: Lunch; Dinner

Price: \$4 - \$13 Features: outdoor seating;

## Okrepcevalnica Sarajevo

[Visit Restaurant Page](#)

Gospodsvetska cesta 43c, Maribor 2000 Slovenia 📞 +386 2 252 34 31

Cuisines: Barbecue; European Meals: Lunch; Dinner Price: \$2 - \$11 Features: outdoor seating;

# Argumentation Search · CERN Search Applications

## Argumentation Search (WEBIS)

args.me extended with OWI: pro & contra arguments on 31 controversial topics; argument quality scoring, key point matching, temporal trend analysis

## CERN Search Applications

- **Institutional**: 25K+ pages crawled from CERN internal web; full OWS pipeline → MOSAIC index for AccGPT knowledge base
- **Nooon**: disability knowledge search (2.59M docs) for HR & Diversity & Inclusion





args

We should abolish the right to keep and bear arms

violence

All

Search

520 results46 ms

Pro-Arguments

ranking score: 9.051source

Argumentative Segment

Since the start of the year, there have been 80 mass shootings and more than 3,300 deaths due to gun violence nationwide, according to the Gun Violence Archive. We don't need any more thoughts and prayers from elected leaders. We shouldn't have any more moments of silence or vigils. We need action.

Key PointQuality Score: 0.880

Gun ownership allows for mass-shootings/general gun violence

ranking score: 8.636source

Argumentative Segment

"Any successful strategy to reduce gun violence requires preventing the diversion of lawful firearms into unlawful commerce," Steven Dettelbach said at the time. "Once there, these firearms end up in the hands of people who are sometimes violent criminals and intend to do harm to the people with whom we live, the innocent people who are victims and survivors of gun violence."

Key PointQuality Score: -0.690

Guns can fall into the wrong hands

Con-Arguments

ranking score: 11.986source

Argumentative Segment

Councilman Johnson, repeat after me: Violence is violent. The number one cause of violence is violent people. While semantically there is an argument that some people aren't violent until provoked, drunk and/or stoned, exposed to constant violence in entertainment media, video games etc., the fact remains that violence always, 100% of the time, involves at least one violent person. Blaming guns for violence is like blaming hammers for shoddy home construction.

Key PointQuality Score: 0.660

ranking score: 8.489source

Argumentative Segment

No matter how many times actual, real-life case studies are laid out for Cleveland's elected officials, they refuse to recognize that the only PROVEN method of reducing violence is removing violent persons from circulation, either through imprisonment or by vibrantly empowering the potential victims to remove the violent person when confronted by one.

Key PointQuality Score: 0.870

Gun ownership promotes self protection

Demo: [args.me/owi](https://args.me/owi) · [args.me](https://args.me)

MOSAIC

ows-lb-2.cern.ch/ui/

80%

MOSAIC Search

Search term:disability

Search

Index Info

Geo Filter:

West:

East:

North:

South:

Index:

Language:

Limit:

Keyword:

default / all

disability-index

default / all

English

German

default / 20

10 items

50 items

1,000,000

Search URL: https://ows-lb-2.cern.ch/mosaic/search?q=disability&index=disability-index

Search result for term: "disability"

Index: disability-index

Number of items: 20

Cockrell Hill Social Security Disability Attorneys | OneCLE Texas Attorney Directory

Kraft Dallas, TX Social Security Disability Lawyer with 54 years of experience (214) 999-9999 2777 N Stemmons Fwy Suite 1300 Dallas, TX 75207 Free ConsultationSocial Security Disability and Personal Injury Baylor Law School Show Preview View Website

Metadata: language:eng, word count:4273, index date:2025-08-10 00:13

Locations:

Keywords: Attorney • Attorneys • Social Security Disability • Cockrell Hill • Texas • https://lawyers.onecle.com/lawyers/social-security-disability/texas/cockrell-hill Insurance Coordinator Manual

Partial disability must result from the same condition as the total disability. Proof of partial disability must be submitted within 31 days of the date the employee's total disability period ends. Pa

Metadata: language:eng, word count:20858, index date:2025-08-10 00:51

Locations:

Keywords: https://oklahoma.gov/egid/insurance-coordinators/manual.html

House of Lords/Commons - Joint Committee on the Draft Disability Discrimination Bill - First Report

Disability Rights Commission, DDB 1, para 10.6.1. British Council of Disabled People, DDB 6, para 9. Law Society, DDB 15, p7. 15(a): The Government agrees with the recommendations of the DRTF that MPs

Metadata: language:eng, word count:12308, index date:2025-08-10 00:25

Locations:

Keywords: https://publications.parliament.uk/pa/jt200304/jtselect/jtdisab/82/8222.htm

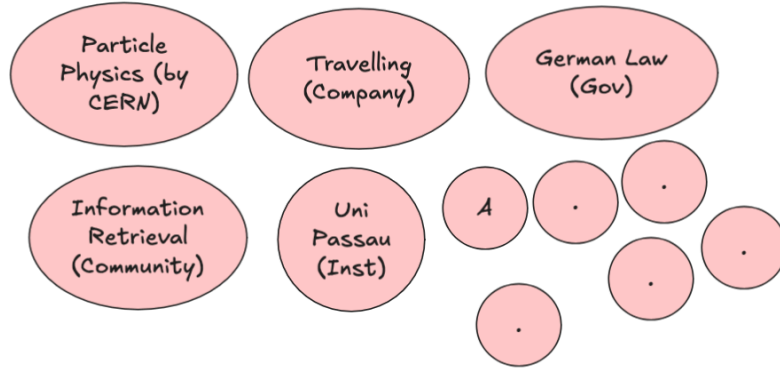
UKRI diversity data for funding applicants and awardees 2020 to 21 update – UKRI

After removing this opportunity, the award rate for fellows reporting a disability was 22% compared with

## Third-Party Projects (FSTPs)

- **AKSE** — argumentation knowledge graph for structured deliberation
- **VERITAS / DEXAI** — verification-oriented RAG for sensitive geopolitical contexts (Ukraine)
- **OMMS** — enriching OpenStreetMap POI with OWI structured data
- **DTCommerce** — product extraction & enrichment pipeline for SME e-commerce
- **CIFFIL** — cross-collection index interoperability for municipal search (Dutch pilots)
- **TILDE** — trustworthy health search with bias detection and entity-level retrieval

# Towards Federated RAG Systems



- RAG / Agentic systems require focused, high-quality collections
- Web information redundancy increases due to LLMs — no one-size-fits-all retrieval
- OpenWebSearch.eu supports low-cost creation of topic-specific RAG systems

## Key ideas:

- Retrieve from well-designed RAG systems instead of generic search engines
- Topic-specific, deduplicated and curated content
- Domain-optimized ranking, aggregation and summarization
- Lightweight browser-based federation with personal data integration

**OwlerLite** (WWW 2026) — privacy-respecting browser extension for scope- and freshness-aware retrieval: user-defined **scopes** (trusted sources), semantic

freshness detection via SimHash + vector similarity, Scope Fidelity improved from 0.64 → **0.83** (negligible NDCG loss)

# Using the Open Web Index — Data & Analytics

# Using the Open Web Index — Data & Analytics

## Published Datasets

| Dataset         | Scale                 | Use case                        |
|-----------------|-----------------------|---------------------------------|
| WebFAQ          | 96M Q&A, 75 langs     | QA training, cross-lingual IR   |
| WebFAQ 2.0      | 198M pairs, 108 langs | Dense retrieval, hard negatives |
| Imprints        | 5.25M hosts, geocoded | Enterprise analytics, Eurostat  |
| German Commons  | 154B tokens           | German LLM pre-training         |
| OWS-Curlie-2025 | 20.7M docs + qrels    | IR evaluation benchmark         |

All datasets available at [openwebindex.eu/corpora](https://openwebindex.eu/corpora)

## Analytics Potential

- **Official Statistics:** enterprise data extraction from legal imprints (Eurostat, Destatis)
- **Media Monitoring:** supply chain analysis, news monitoring (JRC)
- **AI Training Data:** domain-specific corpora for LLM fine-tuning
- **Evaluation Benchmarks:** web-scale IR test collections with relevance assessments
- **Trend Analysis:** temporal web monitoring at Petabyte scale

The OWI is not just for search — it is a data infrastructure for the web-scale analytics ecosystem

# Published Datasets

# WebFAQ — Multilingual FAQ from the Web

- **96 million** Q&A pairs across **75+ languages**
- Extracted from structured FAQ markup (JSON-LD, Microdata) in the Open Web Index
- Fields: question, answer, URL, origin, topic, question type
- Parquet format, hosted on Hugging Face

**Use cases:** multilingual QA training, information retrieval, cross-lingual transfer learning

[huggingface.co/datasets/PaDaS-Lab/webfaq](https://huggingface.co/datasets/PaDaS-Lab/webfaq)

| Top languages | Pairs     |
|---------------|-----------|
| English       | ~30M      |
| German        | ~8M       |
| French        | ~6M       |
| Spanish       | ~5M       |
| ...           | 75+ langs |





# WebFAQ 2.0 — Hard Negatives for Dense Retrieval

- **198 million** FAQ-based Q&A pairs across **108 languages**
- **14.3 million** bilingual aligned QA pairs — largest FAQ-based resource
- **1.25 million** queries with mined hard negatives (20 languages)
- 200 cross-encoder scored negatives per query

## Key contributions:

1. Novel web crawling for diverse content extraction
2. Hard negatives for dense retriever training
3. Contrastive Learning & Knowledge Distillation
4. Open datasets and training scripts

---

*Dinzinger, Caspari, Salman, Topi, Mitrović & Granitzer. "WebFAQ 2.0: A Multilingual QA Dataset with Mined Hard Negatives for Dense Retrieval." arXiv:2602.17327, 2025.*

# Imprints Dataset — Geospatially Grounded Web Curation

- **5.25 million** hosts, **47.8 GB**
- Each website annotated along two dimensions:
  - **Topical classification** via LLM (Foursquare category hierarchy)
  - **Geographic grounding** from legal imprints via geocoding
- **3.08 million** websites geocoded from extracted addresses
- Page types: main page, imprint, contact, about us, privacy, terms

| Country     | Geocoded  | Share |
|-------------|-----------|-------|
| Germany     | 2,636,913 | 85.5% |
| Austria     | 223,805   | 7.3%  |
| Switzerland | 160,683   | 5.2%  |

| Top-Level Topic         | Share |
|-------------------------|-------|
| Business & Professional | 41.6% |
| Community & Government  | 16.2% |
| Arts & Entertainment    | 12.2% |

# German Commons — 154B Tokens for German LLMs

- 154.56 billion tokens, 35.78 million documents
- Aggregated from 41 diverse sources under open licenses
- Quality-filtered, deduplicated, OCR scored

| Domain             | Tokens |
|--------------------|--------|
| News Commons       | 72.67B |
| Cultural Commons   | 54.49B |
| Web Commons        | 19.89B |
| Political Commons  | 3.57B  |
| Legal Commons      | 2.99B  |
| Scientific Commons | 0.84B  |
| Economics Commons  | 0.11B  |

Gienapp, Schröder, Schweter et al. "The German Commons — 154 Billion Tokens of Openly Licensed Text for German Language Models." arXiv:2510.13996, 2025.



# OWS-Curlie-2025 — Web Search Test Collection

- **20.7 million** documents from 6 months of OWI crawl data (March–August 2025)
- English documents from the Curlie subcollection (websites labeled in the Curlie directory)
- Three evaluation splits with student-provided topics and relevance assessments
- Jina V3 embeddings, ClFF indexes for sparse retrieval
- Accessible via owilix CLI and ir-datasets-owi Python package

| Split       | Documents  | Embeddings | Qrels |
|-------------|------------|------------|-------|
| all         | 20,746,181 | --         | --    |
| radboud-val | 63,639     | yes        | yes   |
| radboud     | 90,852     | yes        | --    |
| jena-kassel | 77,382     | yes        | --    |

[openwebindex.eu/corpora](https://openwebindex.eu/corpora) · License: OWIL v1.0



# Using the Open Web Index — Ideas

# Potential Application Domains

Some application scenarios discussed at the start of OpenWebSearch.eu are now in reach:

## Web-Analytics for Official Statistics

**Eurostat:** enterprise information extraction from legal imprints — who is doing business where?



**Destatis:** statistical data extraction from web sources to complement national statistics



## Media Monitoring & Intelligence

**JRC Media Monitor:** monitoring media ecosystems and supply chain signals at European scale (*in start phase*)



## A Live Use Case: EU Toy Safety & the Open Web Index

Can the OWI be used for EU product market surveillance? In preparation of the talk I ran a live exploratory analysis using the OUR<sup>2</sup>S structured-data API plus some AI Agents.

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*"Make an analysis of the OURS API for the OWI corpus: find children's toys and assess how they appear in the dataset — particularly those potentially violating EU regulations."*

Prompt used in JetBrains's Junie AI System with Gemini 3.1 Flash



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Prompt used in Jetbrain's Junie AI System with Gemini 3.1 Flash

## The corpus

| Schema              | Documents |
|---------------------|-----------|
| Web pages           | 310 M     |
| Structured entities | 137.5 M   |
| Unique hosts        | 10 M      |

## Entities Found

- **27 M Products** — largest single type
- 26 M Organisations
- 18 M Articles
- 3.8 M Local Businesses
- 2.8 M Events

→ Products are the single largest entity type — directly usable for market analysis.

# A Live Use Case: Products at Scale — Toys in the Index

## Pricing & geography

- ~15 M entities carry price data
- **3 M EUR-priced** products → significant EU market coverage
- Also: USD 2.2 M · UAH 1.9 M · BRL 1.2 M

## Toy query results (live)

- "children toy" → **102,530 products**
- With pricing → **51,384** (50 % have structured prices)
- "Spielzeug Kinder" → **7,888** German results

## Sample items found

- *"1:24 die-cast metal sports car"* — \$90.99
- *"Kids Microscope 1000× Magnifier"* — \$59.00
- *"48-teiliges Einhorn Prinzessin Set"* — €8.44

## Structured fields available per product

- Name, brand, GTIN/EAN
- Price, currency, availability
- Description, images, ratings

~50 % of toy products include structured price data, reflecting real e-commerce markup adoption.

~92 % of all entities **lack location data** — a significant data-quality finding.

Whole analysis took 20 minutes in parallel to working on this presentation.

## A Live Use Case: Conclusion

### What the OWI provides today

- Structured Microdata with some features
  - GTIN/EAN presence assessment
  - Brand & manufacturer data
- Completeness and parsing ok, but should be improved

### What would be missing for full EU compliance

- CE marking (*not in schema.org*)
- Age suitability / safety warnings
- Material composition
- ...

Could be added to parser

The OWI is a **foundation for such a use case** — Without an aggregated web index, such an analysis would not be possible. Of course you can use commercial search engines for such tasks, but they are not transparent and you don't know what data they use.

# Other Research Directions

## Next-Generation Retrieval

- Federated RAG across distributed OWI slices and 3rd-party corpora
- Agentic search orchestration for autonomous knowledge discovery
- Multimodal indexing: images, tables, charts alongside text
- Temporal web search: tracking how the web changes over time

## AI & LLM Infrastructure

- Domain-specific corpora for sovereign European LLM training
- Retrieval-Augmented Generation benchmarks on OWI data
- Open evaluation infrastructure for IR and NLP research
- AI Agents & Search - actively pursued by Uni Passau
- **Multi-loop LLM orchestration:** classify incoming tasks, represent task types as vectors, and use cross-encoder-style scoring to route or sequence specialised LLM loops for the identified tasks
- **Stacked Kohonen embeddings:** map each sentence token to the top-n matching SOM codebook vectors, use their grid positions as explicit embedding coordinates, and condition the next token's SOM lookup on the previous token's match to form a joint sequence vector
- **Capability allocation across parameter space:** which capabilities — knowledge, reasoning, behaviour — should live in *model parameters* vs. *indexed memory* vs. *interaction traces*? Open design question; envisioned figure maps capability types across the parameter / retrieval / trace space (parked from the TPC26 "Open System Question" slide)



## Societal & Policy Applications

- **Information quality monitoring:** detecting coordinated inauthentic behaviour, SEO spam
- **Digital literacy tooling:** transparent, inspectable search for education
- **Regulatory compliance:** evidence-based auditing of search engine behaviour
- **Digital Sovereignty:** European alternative to US-dominated search infrastructure

The OWI is an open infrastructure — we invite researchers, businesses, and public institutions to build on it

We are looking to be part of interesting research projects to advance the OWI.

We hope to build an open ecosystem that puts open, fair and unbiased information exchange and societal values before commercial interest.

# Conclusion

# In Summary

## What we built

- A **federated, open web index** — 1.1 PB raw data, 810+ daily datasets, running 24/7 across European HPC centres
- A **full open-source pipeline** — crawl → preprocess → index → distribute, all reproducible
- A **growing ecosystem** — search apps, RAG systems, datasets, and analytics tools built on the OWI

## Why it matters

- Web data and web search are **critical infrastructure for Europe's digital sovereignty**
- The OWI **reduces upfront costs** for tapping the web as resource
- Open, transparent, and FAIR — enabling **reproducible research** and **open innovation**
- **Collaborative** approach to build and maintain the infrastructure

OpenWebSearch.eu proves that a European open web index is technically feasible, legally navigable, and economically viable

# Before we come to the Questions



# We are looking for ...

## Helping Hands

- **Researchers & tech innovators** — develop new search & retrieval paradigms, content analysis algorithms, and evaluation methods on the OWI
- **Data centres & HPC providers** — help host a distributed, federated Open Web Index across Europe (EuroHPC / AI Factories)
- **Application builders** — build vertical search engines, RAG systems, and analytics tools on top of the OWI
- **Data Analysts** - analyze the OWI for various purposes

## Helping Funds

- **Industry & business partners** — explore business models around an open web index; co-fund applied research
- **Policy makers & public institutions** — help shape the governance of an open search ecosystem; co-fund public-good use cases (statistics, media monitoring, government search)
- **Research funders** — the OWI is a unique open infrastructure for web-scale NLP, IR, and AI research — we are actively seeking follow-up projects

## Get in touch

|                        |                                                                                                                 |
|------------------------|-----------------------------------------------------------------------------------------------------------------|
|                        |                                                                                                                 |
| <b>Website</b>         | <a href="https://openwebsearch.eu">openwebsearch.eu</a>                                                         |
| <b>Dashboard</b>       | <a href="https://openwebindex.eu">openwebindex.eu</a>                                                           |
| <b>Code Repository</b> | <a href="https://opencode.it4i.eu/openwebsearcheu-public/">https://opencode.it4i.eu/openwebsearcheu-public/</a> |
| <b>Community</b>       | <a href="https://openwebsearch.eu/community">openwebsearch.eu/community</a>                                     |
| <b>Join</b>            | <a href="mailto:join@openwebsearch.org">join@openwebsearch.org</a>                                              |
| <b>General</b>         | <a href="mailto:community@openwebsearch.eu">community@openwebsearch.eu</a>                                      |

Thank you — Questions, Comments, Feedback very welcome!